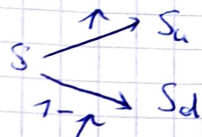


Financial Mathematics: Exercises Lecture 10



$$S = 100, S_u = 150, S_d = 75$$
$$p = \frac{1}{2} \text{ (real-world prob.)}$$
$$r = 0$$

European call, $K = 110$.

• Risk-neutral valuation (RN)

$$q = \frac{e^{r\Delta t} - d}{u - d}$$

$$S_u = 150 = uS = u100 \Rightarrow u = 1.5$$
$$S_d = 75 = dS = d100 \Rightarrow d = 0.75$$

$$\Rightarrow q = \frac{1 - 0.75}{1.5 - 0.75} = \frac{1}{3}$$

$$\text{So, the price is: } C^{RN} = \frac{1}{1+r} (qC_u + (1-q)C_d)$$

$$C_u = \max(S_u - K, 0)$$
$$= 40$$

$$C_d = \max(S_d - K, 0)$$
$$= 0$$

$$\Rightarrow C^{RN} = \frac{1}{1} \left(\frac{1}{3} \cdot 40 + \frac{2}{3} \cdot 0 \right)$$
$$= \frac{40}{3} = 13.33$$

• Real-world (RW):

$$C^{RW} = \frac{1}{1+r} (p \cdot C_u + (1-p) \cdot C_d)$$
$$= \frac{1}{1} \left(\frac{1}{2} \cdot 40 + \frac{1}{2} \cdot 0 \right)$$
$$= 20$$

$$\Rightarrow C^{RN} \neq C^{RW}$$

• Arbitrage opportunity:

$$\Delta^* = \frac{C_u - C_d}{S_u - S_d} = \frac{40 - 0}{150 - 75} = \frac{40}{75} = \frac{8}{15}$$

$$\alpha^* = \frac{1}{1+r} (C_u - \Delta^* S_u) = 40 - \frac{8}{15} \cdot 150 = 40 - 80 = -40$$

$$\Rightarrow H^* = \alpha^* + \Delta^* S \text{ replicates } C^{RN}$$

~~(Borrow 40)~~

↳ replicating portfolio, worth 13.33 ($= C^{RN}$)

Arbitrage strategy:

At $t=0$:

→ (short position)
• Sell the option

• Buy H^*
(sell again at maturity)

CF now	CF at maturity	
	if S_u	if S_d
+20	$-(S_u - K)$ ($= -40$)	0
-13,33	$\alpha^* + \Delta^* S_u$ ($= 40$)	$\alpha^* + \Delta^* S_d$ ($= 0$)
Net: +6,67	Net: 0	Net: 0

(Remark: "Buy H^* " means:

• borrow 40 ($= -\alpha^*$) (at $r=0$)

• buy Δ^* shares of S ($\Delta^* S = -53,33$)

So you ~~have~~ make a riskless profit now.